

B — Reproductive Physiology

This is a separate revision topic under the SAI syllabus area **Human Physiology: “Endocrine and reproductive systems.”** It is useful for understanding growth, puberty, menstrual function, energy availability and athlete health. It does not mean that every athlete should be judged by the same sex-based or cycle-based rule.

The control pathway

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Hypothalamus
  ↓ releases gonadotropin-releasing hormone
Anterior pituitary gland
  ↓ releases follicle-stimulating hormone and luteinizing hormone
Ovaries or testes
  ↓ release reproductive hormones and gametes
Target tissues and reproductive function
```

Most of this pathway uses **negative feedback**: when reproductive hormones rise, they usually reduce further stimulation from the brain and pituitary gland. A special exception occurs before ovulation, when sustained high estrogen changes to **positive feedback** and produces a sharp luteinizing-hormone surge.

High-yield hormone map

Hormone or structure	Main exam-relevant role
Follicle-stimulating hormone	Supports ovarian follicle development; in testes, acts mainly on Sertoli cells to support sperm production
Luteinizing hormone	Triggers ovulation; in testes, stimulates Leydig cells to produce testosterone
Estrogen	Supports female reproductive development and helps rebuild the uterine lining
Progesterone	Stabilises and prepares the uterine lining after ovulation and supports pregnancy
Testosterone	Supports male reproductive development, sperm production and anabolic tissues
Inhibin	Selectively reduces follicle-stimulating-hormone release when the gonads are producing gametes
Human chorionic gonadotropin	Early-pregnancy hormone that helps maintain progesterone production until the placenta takes over much of that role

Menstrual-cycle sequence

The cycle is variable; “28 days” is a teaching example, not a rule for every person.

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Menstruation → follicular phase → ovulation → luteal phase
  low hormones      estrogen rises      luteinizing-      progesterone rises
  lining sheds      follicle develops    hormone surge    corpus luteum forms
                                ↓
                        no pregnancy: hormones fall → next menstruation
```

- **Menstruation:** the uterine lining is shed after estrogen and progesterone fall.
- **Follicular phase:** a follicle develops in the ovary and estrogen generally rises.
- **Ovulation:** the luteinizing-hormone surge releases the oocyte (egg cell).
- **Luteal phase:** the corpus luteum produces progesterone; if pregnancy does not occur, it regresses and hormone levels fall.

Male reproductive physiology

- The **testes** produce sperm and testosterone.
- **Sertoli cells** support developing sperm; **Leydig cells** produce testosterone when stimulated by luteinizing hormone.
- Sperm production requires coordinated follicle-stimulating-hormone action, testosterone and suitable testicular temperature.

Puberty, maturation and key terms

- **Puberty:** activation of the brain–pituitary–gonad pathway, producing reproductive maturity and secondary sexual characteristics.
- **Menarche:** the first menstrual bleeding.
- **Amenorrhea:** absence of menstruation. In an athlete, it should not automatically be dismissed as a normal training effect.
- **Ovulation:** release of an oocyte from an ovary.
- **Spermatogenesis:** production of sperm cells.
- **Gametogenesis:** production of reproductive cells—sperm or oocytes.

Exercise and athlete health

- Training response varies between individuals. Do not prescribe a universal “best” training week solely from the menstrual-cycle phase; consider symptoms, performance, recovery, medical status and the athlete’s preferences.
- **Low energy availability** means that food energy left for normal body functions is inadequate after exercise expenditure is considered. It can disturb reproductive hormones and menstrual function, impair bone health and recovery, and contribute to **Relative Energy Deficiency in Sport**.
- Repeated missed periods, cycle changes, reduced libido, persistent fatigue, stress injuries or unexplained performance decline deserve confidential assessment by the appropriate medical team. They are not proof of poor commitment or a reason to increase training automatically.
- In adolescents, biological maturation can change size, strength and reproductive function at different rates. Selection and training decisions should therefore use growth and health monitoring, not appearance alone.

Memory tactics

- **“Brain → pituitary → gonad → target.”** Follow the pathway from the control centre to the final organ.
- **“Follicle-stimulating hormone supports; luteinizing hormone launches.”** The first supports follicles/sperm production; the second triggers ovulation or stimulates testosterone production.
- **“Estrogen builds; progesterone prepares.”** Estrogen generally rebuilds the uterine lining; progesterone makes it suitable for support after ovulation.
- For the cycle, remember **“shed, grow, release, support”**: menstruation, follicular growth, ovulation, luteal support.
- For an athlete with menstrual disturbance, ask **“fuel, load, health”**: is energy intake adequate, is training stress excessive, and is another medical cause possible?

Quick self-test

1. Which gland releases follicle-stimulating hormone and luteinizing hormone?
2. What event is triggered by the mid-cycle luteinizing-hormone surge?
3. Which hormone is dominant in the luteal phase?
4. Why should repeated missed periods in an athlete be assessed rather than dismissed?

Sources
